

RELAY SETTING SHEET — TRANSFORMER PROTECTION

Relay: **SEL-787** · Tag: T1-DIFF-87 · Setting Group: 1 · Project: synthetic example · Engineer: J. Engineer P.E.

Element	Function	Setting	Units	Curve
87T	Differential	0.30	pu	—
87HS	High-Set Differential	8.0	pu	—
50P1	Phase Overcurrent	1200	A	—
50P2	Phase Overcurrent	3600	A	—
51P	Phase Time-OC	600	A	U2 (IEC VI)
51P TD	Time Dial	0.55	—	—
50N	Neutral Overcurrent	150	A	—
51N	Neutral Time-OC	75	A	U2 (IEC VI)
51N TD	Time Dial	0.40	—	—
27P	Phase Undervoltage	0.85	pu	—
59P	Phase Overvoltage	1.15	pu	—
81U	Underfrequency	59.5	Hz	—
81O	Overfrequency	60.5	Hz	—

TRIP TARGETS

TRIP1 = 87T + 87HS
TRIP2 = 51P + 51N
TRIP3 = 27P + 59P + 81U + 81O
ALARM = 50P1 (instantaneous)

Reference: IEEE C37.91 — Guide for Protecting Power Transformers · IEEE Std 242 §11 — Relay coordination